Project Summary: Liver Screening Data Analysis

Executive Summary

This report summarizes the key findings from an exploratory data analysis (EDA) on liver screening data. The analysis aimed to understand the characteristics of the patient population, identify patterns in liver function test results, and explore potential differences between demographic groups. The insights gained will inform strategies for targeted screening and intervention programs.

Key Findings

Based on the analysis of **583** patient records, the following key findings were identified:

Demographics

The dataset is predominantly composed of **Middle-aged adults (40-59)**, representing the largest age group in the study. There is a higher representation of **males** compared to females in the dataset.

Liver Function Test Abnormalities

A significant portion of the dataset shows **elevated levels** in key liver function tests, particularly **ALP, ALT, and AST**. This indicates a high prevalence of potential liver function abnormalities within the screened population.
Low levels of **Albumin** and **A/G Ratio** are also frequently observed, which can be indicative of liver dysfunction or other underlying health issues.

Correlations between Liver Function Tests

Strong positive relationships were observed between **Total Bilirubin and Direct Bilirubin**, and between **Total Proteins and Albumin**. These correlations are expected and align with known physiological relationships.

A positive trend was noted between **AST and ALT**, two important indicators of liver health.

Gender Differences

Statistically significant differences were found in the levels of **Total Bilirubin, ALT, and Albumin** between male and female patients. Further investigation is needed to understand the clinical significance of these differences.

Data Management Highlights

The raw data contained missing values which were successfully imputed using the KNN method.
Duplicate records were identified and removed to ensure data accuracy.
Data types were standardized for consistency.

Implications for Stakeholders and Leadership

The findings suggest a need to focus on the middle-aged male population for targeted liver health initiatives. The high prevalence of elevated liver enzymes and low protein/albumin levels warrants further clinical investigation and potentially more aggressive screening or early intervention programs. Understanding the gender-specific differences in liver function tests can help tailor diagnostic and treatment approaches.

Next Steps

1. Clinical Validation: Collaborate with medical professionals to validate the clinical significance of the identified abnormalities and gender differences.
2. Further Analysis: Investigate the causes of the high prevalence of elevated liver enzymes and low protein/albumin levels.
3. Targeted Programs: Develop and implement targeted screening and intervention programs based on the age and gender insights.